AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 21 · STUDENT LESSON

Good Data, Bad Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Before you start

Week 3: measurements require units and a consistent method.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Check records against a stated format, unit and meaning before using them. Try the recall before opening notes.

Your learning target

I can check records against a stated format, unit and meaning before using them.

I can distinguish an **evidence**-based **correction** from an unsupported guess.

I can keep original values and a **change log**, then report what remains unresolved.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Confidence Is Not **Correctness**. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Save New Examples for Testing. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Coverage, **correctness**, and **relevance** matter more than raw quantity.



Your target

LEARNING TARGET A table lists leaf lengths 8 cm, 90 mm, blank and 700 cm. Propose a documented cleaning action for each row. Which values can you safely change immediately?

Learner name:

Readiness for the connected explanation

A recommender needs availability, but that field is blank. Can the highest preference score establish that the item is available? Explain.

This week’s end goal

By the end, I can repair data with traceable **evidence** while preserving valid zero and unknown.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Good Data, Bad Data case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



Start with this puzzle

MYSTERY One animal dataset has 1,000 blurry **duplicate** photos; another has 120 clear, correctly labeled photos across settings and animal types.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

A tidy table can still contain wrong information. **Validation** checks whether a record follows a **schema**. A **schema** says which fields are required and what their values mean.

Passing these checks makes a record usable for a stated task; it does not prove that the measurement or claim is true.

Our fictional RampLog records one ramp trial per trial ID. The fields are trial ID, height in centimetres, travel in centimetres and source card. Height must be a whole number from 1 through 10.

Travel must be a whole number from 0 through 10. The source card identifies the original **observation**. Repeated trial IDs need investigation before counting trials.

The purpose is to count trials whose travel is at least 4 cm. A measured zero means no travel; a blank means no value was recorded. Never silently replace a blank by zero.

Use only the invented records here; no personal information is needed.

STUDENT LESSON · 3 OF 18

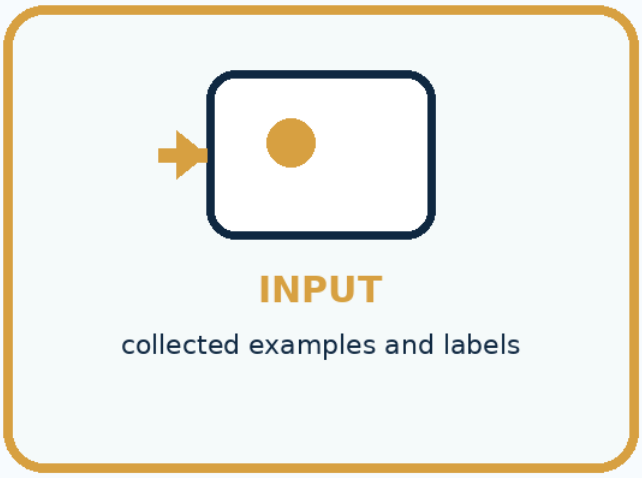
See how it works

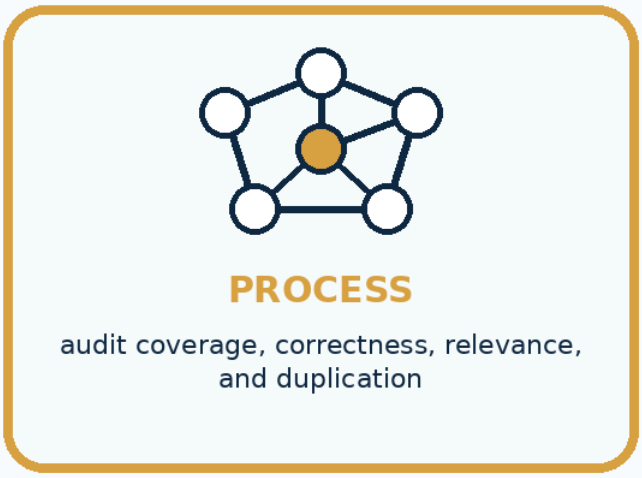
> Follow the mechanism for Good Data, Bad Data. Connect each labelled input, deciding step and result.

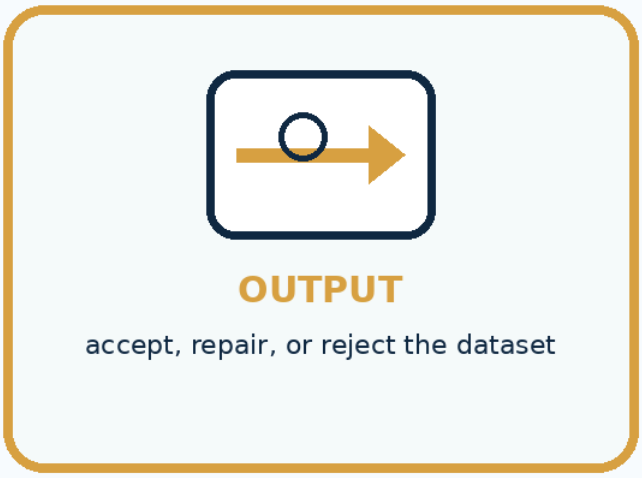
VISUAL MECHANISM

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT collected examples and labels

PROCESS audit **coverage**, **correctness**, **relevance**, and duplication

OUTPUT accept, repair, or reject the dataset

Key idea

CORE IDEA **Coverage**, **correctness**, and **relevance** matter more than raw quantity.

How the mechanism works

UNDER THE HOOD Data quality is multidimensional. More rows can amplify **duplicate**s, systematic labeling errors, stale conditions, or irrelevant fields.

Fitness depends on the intended population and decision, not a universal quality score.

Draw a fresh example with all three stages:

A closer explanation

Preserve the original and explain each decision

ORIGINAL RECORD	EVIDENCE CHECK	WORKING DECISION
T2 height 60 cm	Card C2 says 6 cm	Correct to 6; log C2
T3 travel blank	Card C3 also blank	Keep missing; flag
T4 travel 0 cm	Card C4 confirms zero	Keep measured zero

C2 supports correcting T2. C3 cannot supply a missing measurement. C4 shows that zero is a recorded observation, not a placeholder for missing data.

The complete raw table is T1: height 4, travel 3, card C1; T2: height 60, travel 5, card C2; T3: height 5, travel blank, card C3; T4: height 6, travel 0, card C4. A second export row repeats T1 with height 4, travel 3 and C1.

C1 confirms one trial T1, not two. C2 reads height 6 and travel 5. C3 has no travel measurement. C4 reads height 6 and travel 0.

Work in a separate corrected copy. Record T2 height 60 to 6, **evidence** C2, transcription error, reviewer and review date. Link the repeated T1 export row to the first T1; count that single trial once and keep both raw rows.

Do not remove two different trials merely because their measured numbers match.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

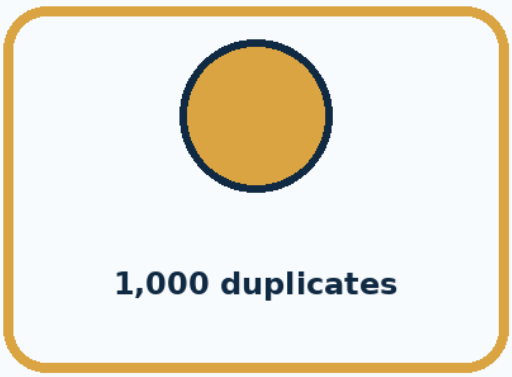
Compare the cases

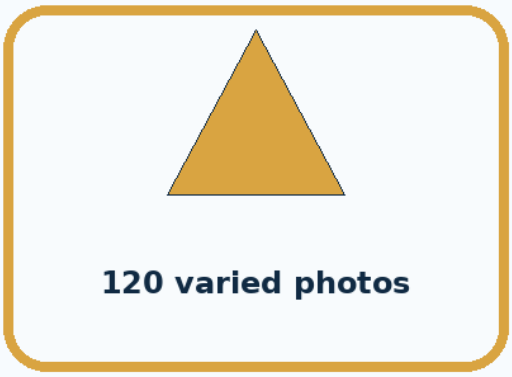
> Compare the supplied cases for Good Data, Bad Data. Identify the detail that changes the conclusion.

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









The case to investigate

CASE 1 — 1,000 **duplicate**s Clue: high count, low variety. Meaning: weak **coverage**.

The case to investigate

CASE 2 — 120 varied photos Clue: settings and types represented. Meaning: stronger **coverage**.

The case to investigate

CASE 3 — Wrong labels Clue: targets conflict with reality. Meaning: **correctness** failure.

The case to investigate

CASE 4 — Owner birthday Clue: unrelated to animal type. Meaning: irrelevant/private.

Keep this boundary clear

BOUNDARY Large quantity cannot compensate for systematic absence or incorrect labels.

Which two pictures are easiest to confuse? What **evidence** separates them?

Change one thing in the pictured case

Change one thing

Change: The source log gives B only as bare 200 with no unit.

Predicted result

Prediction: B remains unresolved rather than becoming 20 cm.

Why the result follows

Reason: Without a confirmed unit, division by 10 would be an unsupported assumption.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Good Data, Bad Data to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Word	Meaning	My example
coverage	how well examples represent needed situations	
correctness	whether records and labels are accurate	

Word	Meaning	My example
relevance	whether data supports the task	
duplicate	repeated record adding little new coverage	

Misconception detective

> **Find the unsupported claim**
>
> CLAIM TO REPAIR The bigger dataset must be better.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

Word	Meaning
Schema	A description of required fields, formats, units and allowed values
Validation	Checking a record against stated requirements
Provenance	A record of where information came from
Correction	A change supported by evidence about the intended value
Change log	A record of what changed, why, when and on what evidence

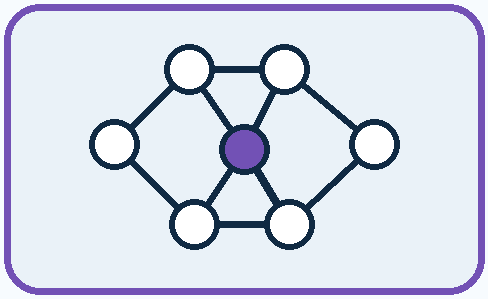
Follow a worked example

> Follow the worked trace for Good Data, Bad Data, predict a step, then compare with its explanation.

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



The case to investigate

CASE One animal dataset has 1,000 blurry **duplicate** photos; another has 120 clear, correctly labeled photos across settings and animal types.

Step 1

STEP 1 — OBSERVE collected examples and labels

Step 2

STEP 2 — TRACE audit **coverage**, **correctness**, **relevance**, and duplication

Step 3

STEP 3 — CONCLUDE accept, repair, or reject the dataset

Model answer

MODEL REASONING Choose the set that covers intended conditions, uses checked labels, includes relevant fields, and avoids misleading duplication. State remaining gaps rather than declaring any dataset perfect.

A claim to avoid

DO NOT OVERCLAIM Large quantity cannot compensate for systematic absence or incorrect labels.

What new **evidence** could change this answer?

A closer explanation

Problem	Permitted action here	Unsupported shortcut
Invalid height 60	Check C2; correct to 6	Guess by deleting a zero
Missing travel	Keep missing; seek source	Fill with the class average
Repeated T1 ID	Confirm one trial on C1; count once	Treat every repeated number as a duplicate
Two sources disagree	Flag conflict; ask for resolution	Pick the value that improves the result

A conversion changes units while preserving the quantity. If an authorised source records 50 mm, that is 5 cm because 10 mm equals 1 cm. Write the original unit, conversion and resulting unit in the log.

A bare value 50 without a known unit does not justify assuming millimetres.

Worked explanation · Good Data, Bad Data

A plausible value can still be a transcription error: 7 would pass the height rule even if its source said 6. Conversely, an unusual but supported value is not automatically wrong.

Checking format and checking agreement with a source are different jobs. A source can itself contain an error, so retain its identity and any unresolved concern.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Good Data, Bad Data

A fictional toy-car distance table needs cleaning before use.

The facts we will use

Supplied facts

Contract: unique run ID; whole distance 0–30 cm. Raw rows: A=0 cm, B=200 mm, C=blank, another A=0 cm.

Supplied facts

Source log confirms one A run and B's original measurement of 200 mm; C has no measurement.

AI ACADEMY / AI-1/21

Good Data, Bad Data

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Validate rows

A's zero is valid; B has wrong unit; C is missing; A is duplicated.

2. Evidence repair

200 mm / 10 =20 cm; source log supports retaining one A.

3. Keep unknown

C stays unresolved; do not replace it with 0.

4. Clean record

Usable unique rows: A=0 cm, B=20 cm; retain a separate change log.

CHANGE ONE THING

The source log gives B only as bare 200 with no unit.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Validate rows	A’s zero is valid; B has wrong unit; C is missing; A is duplicated.
2. Evidence repair	200 mm ÷10 =20 cm; source log supports retaining one A.
3. Keep unknown	C stays unresolved; do not replace it with 0.
4. Clean record	Usable unique rows: A=0 cm, B=20 cm; retain a separate change log.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Cleaning is not permission to make inconvenient values plausible. The source log supports a unit conversion and removal of a **duplicate** copy. Zero means a measured zero distance; blank means no supplied measurement.

Keep the raw table and explain every change so someone can reconstruct what happened.

What this example does not establish

Keep the stated limit

A clean table can still contain measurement or labelling errors that these checks do not detect.

Keep the stated limit

Read the labelled picture: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?

Keep the stated limit

Change one thing: The source log gives B only as bare 200 with no unit. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Good Data, Bad Data claim and explain a repair using the supplied evidence.

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
1,000 duplicates	high count, low variety	weak coverage
120 varied photos	settings and types represented	stronger coverage
Wrong labels	targets conflict with reality	correctness failure
Owner birthday	unrelated to animal type	irrelevant/private

Repair two tempting claims

Challenge claim A

CLAIM A The bigger dataset must be better.

Repair:

Challenge claim B

CLAIM B Clean formatting means the data is correct.

Repair:

A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

A closer explanation

Check the claim against the facts

After changing a value, run the **schema** checks again and compare the corrected copy with the raw copy and log. Recount distinct trials.

> Check that every **correction** has **evidence** and that zero, missing and conflict remain separate. A repair can introduce a new mistake if a value is changed in the wrong row.

The corrected RampLog supports a statement about its three assessed trials. It does not establish how every ramp or future trial behaves. Better formatting cannot create missing **observation**s or broaden what was sampled.

Next week asks which cases a dataset actually covers.

> **A table now has no blank cells because a designer guessed all missing measurements. Explain why this does not establish better evidence, and state what a trustworthy repair record should retain**

STUDENT LESSON · 8 OF 18

Practise together

> Practise Good Data, Bad Data with support, then explain the deciding step in your own words.

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: **Coverage**, **correctness**, and **relevance** matter more than raw quantity.

Use a fresh example to explain **coverage**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Clean formatting means the data is correct.

Invent one edge case and name the safe response.

Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

After the documented **correction** and confirmed **duplicate** handling, RampLog contains four distinct trials: T1, T2, T3 and T4. Three have recorded travel. T1 has 3 cm, T2 has 5 cm and T4 has 0 cm.

Only T2 meets at least 4 cm, so the result is 1 of 3 assessed trials, with 1 trial missing travel. Do not report T3 as a failure or hide it from the account.

> **A reviewer fills T3 travel with 4 because it makes the results look more balanced. Identify the unsupported change, write an honest report using the recorded evidence, and name the source information still needed**

Describe which records entered the calculation and why. Keep unresolved records visible in a separate status list.

A new measurement is a new **observation** with its own time and identity; it should not be passed off as the missing **observation** from an earlier trial.

Practise with less help: Write a traceable correction

Use workbook W2 for the connected example “Data quality and documented repair”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Data quality and documented repair. Proposed explanation: “Every valid value is true”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Which repairs are **evidence**-backed in the original case, and which row must remain unknown?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

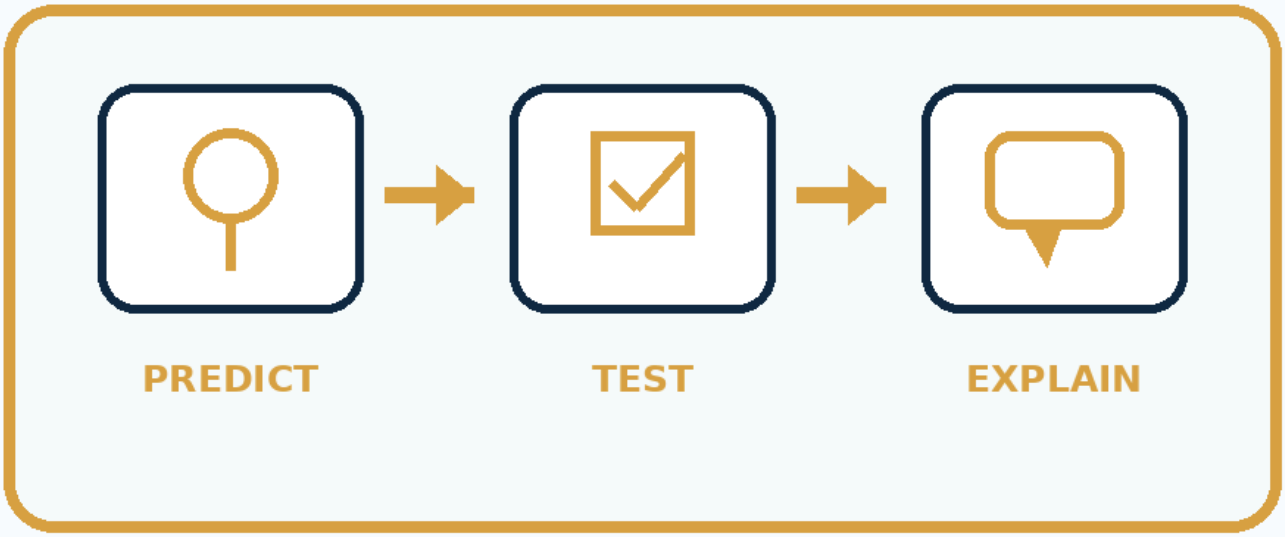
Run the investigation

> Investigate Good Data, Bad Data. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

Dataset Quality Audit

Predict first, test honestly, and explain what the **evidence** means.



1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Audit two mini datasets and choose the safer one.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

Case	Prediction	Observed result	What it means
Normal			
Edge			

Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

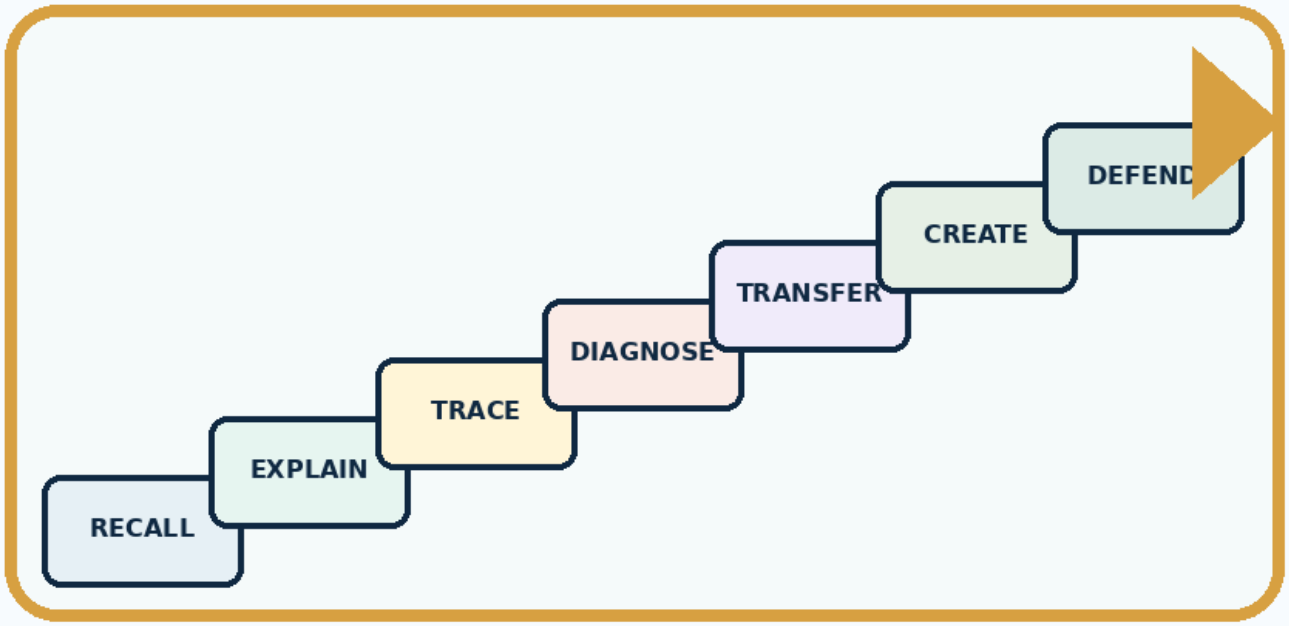
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Good Data, Bad Data. Explain why each step is justified.

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **coverage**.

EXPLAIN **Coverage**, **correctness**, and **relevance** matter more than raw quantity.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: The bigger dataset must be better.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

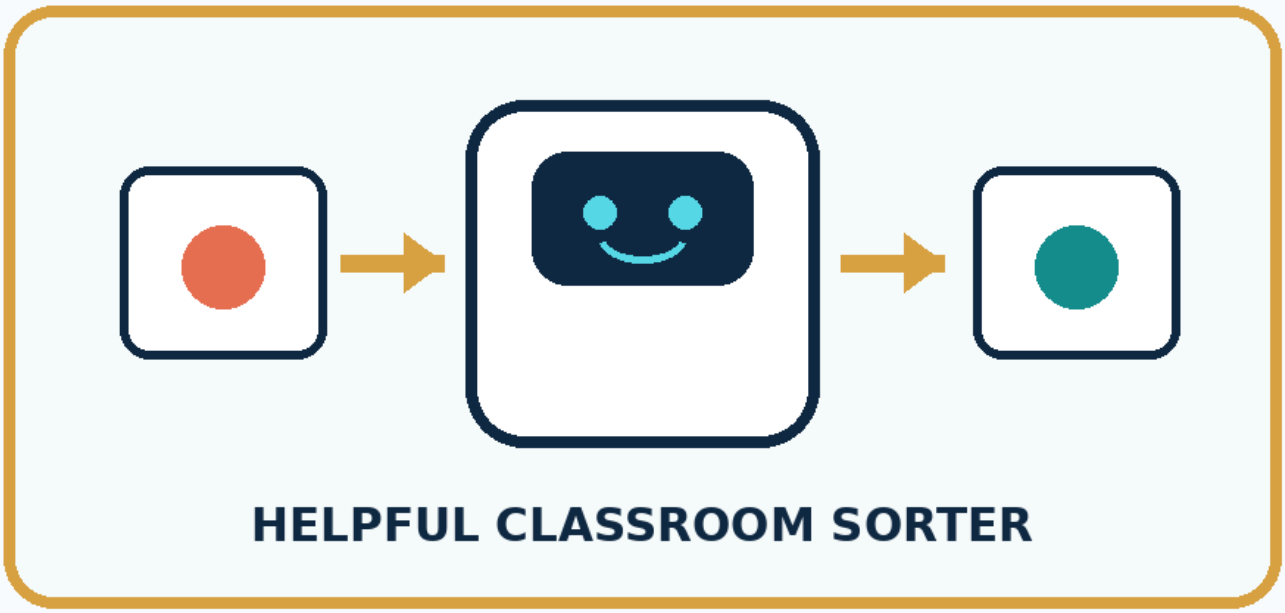
Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



THIS WEEK'S CHECKPOINT Audit the sorter's example cards and add a reason for including each field and condition.

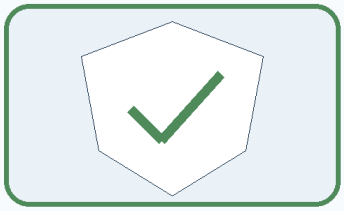
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



Responsible design

RESPONSIBLE DESIGN Large quantity cannot compensate for systematic absence or incorrect labels.

My project decision:

Evidence for your project

Keep raw sorter **input**s, validated records and an **evidence**-linked repair log as different artifacts.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.



What success looks like

REQUIRED PROOF Audit two mini datasets using **coverage**, **correctness**, **relevance**, and duplication; choose the safer one with **evidence**.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

Check your explanation

SELF-CHECK Choose the set that covers intended conditions, uses checked labels, includes relevant fields, and avoids misleading duplication. State remaining gaps rather than declaring any dataset perfect.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Missing and Unbalanced Examples.

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Good Data, Bad Data case for supported practice. Repeated examples do not become fresh assessments.

Week 21 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Ordinary pencil lengths are recorded as 12 cm, 14 cm, blank and 130 cm.

The question

> What needs checking? May 130 silently become 13?

Worked explanation

Investigate missing and unusually large values using the source and units. Do not invent a repair. Document verified **correction**s.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Check suspicious data before changing it

Evidence or stage	Value or result
12 cm	Plausible recorded length
14 cm	Plausible recorded length
Blank	Missing measurement
130 cm	Check source and units

Ordinary pencil lengths are recorded as 12 cm, 14 cm, blank and 130 cm.

> What needs checking? May 130 silently become 13?

Investigate missing and unusually large values using the source and units. Do not invent a repair. Document verified **correction**s.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Audit an independent paper-plane log

Each flight ID must be unique; distance is a whole number from 0 through 20 cm. Raw rows: F1 8 cm; F2 120 cm; F3 blank; F4 0 cm; another F1 8 cm. Source S1 confirms one F1 flight. S2 says F2 travelled 12 cm. S3 is blank.

S4 confirms zero. Preserve the raw rows, describe supported repairs and give the number of distinct flights. Report how many assessed flights travelled at least 10 cm, including missing records separately.

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 21 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A table labels a cat photo dog and lists the same photo three times. Identify both data problems and one sensible repair for each.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Audit the sorter's example cards and add a reason for including each field and condition.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A table lists leaf lengths 8 cm, 90 mm, blank and 700 cm. Propose a documented cleaning action for each row. Which values can you safely change immediately?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

Use this map to navigate Good Data, Bad Data, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional toy-car distance table needs cleaning before use.

By the end, I can	How I will show it
Check records against a stated format, unit and meaning before using them	Explain the deciding step, show a trace or test, and state one limit.
Distinguish an evidence-based correction from an unsupported guess	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Keep original values and a change log, then report what remains unresolved	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Good Data, Bad Data without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Check records against a stated format, unit and meaning before using them

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional toy-car distance table needs cleaning before use.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**.

1. Clean record

2. **Evidence** repair

3. Keep unknown

4. Validate rows

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Clean record Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**. Cleaning is not permission to make inconvenient values plausible.

The source log supports a unit conversion and removal of a **duplicate** copy. Zero means a measured zero distance; blank means no supplied measurement.

Keep the raw table and explain every change so someone can reconstruct what happened.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The source log gives B only as bare 200 with no unit. Which conclusion is supported by the supplied facts?

1. B remains unresolved rather than becoming 20 cm.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: B remains unresolved rather than becoming 20 cm. Without a confirmed unit, division by 10 would be an unsupported assumption.

Repair a missing building block

Review Recommendations (ai-1/20). Goal: Apply eligibility requirements before ranking candidates by preferences. Short practice: Why does the recommendation change without changing anyone's preferences?.

Return to this week and explain what you changed.

Review Confidence Is Not **Correctness** (ai-1/18). Goal: Distinguish a confidence claim from **evidence** that a **prediction** was correct. Short practice: Was the unknown label guaranteed correct because its confidence was **80%**?.

Return to this week and explain what you changed.

Review Save New Examples for Testing (ai-1/15). Goal: Separate examples used to choose a model from examples reserved for its final evaluation. Short practice: Does 4/4 on the replay replace the original 3/4 result? Explain..

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

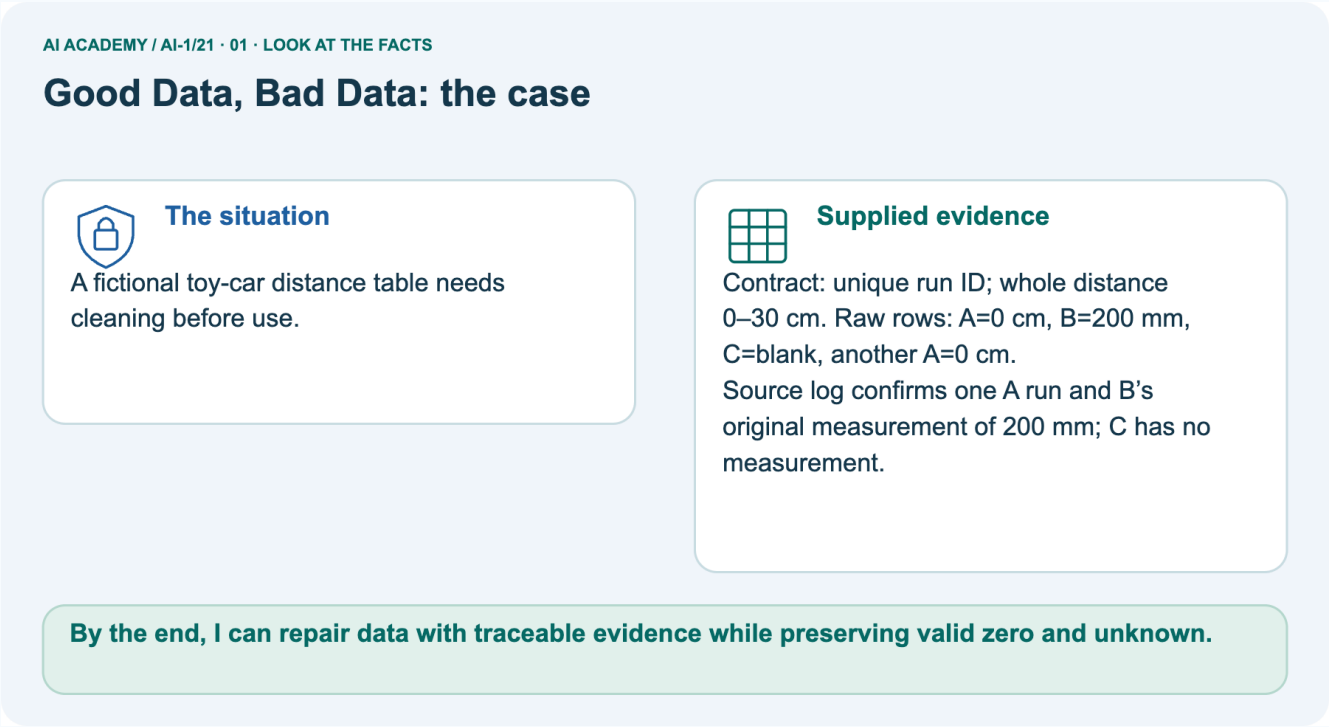
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Good Data, Bad Data graphic using its labels, deciding evidence and relevant limitation.

01. Good Data, Bad Data: the case



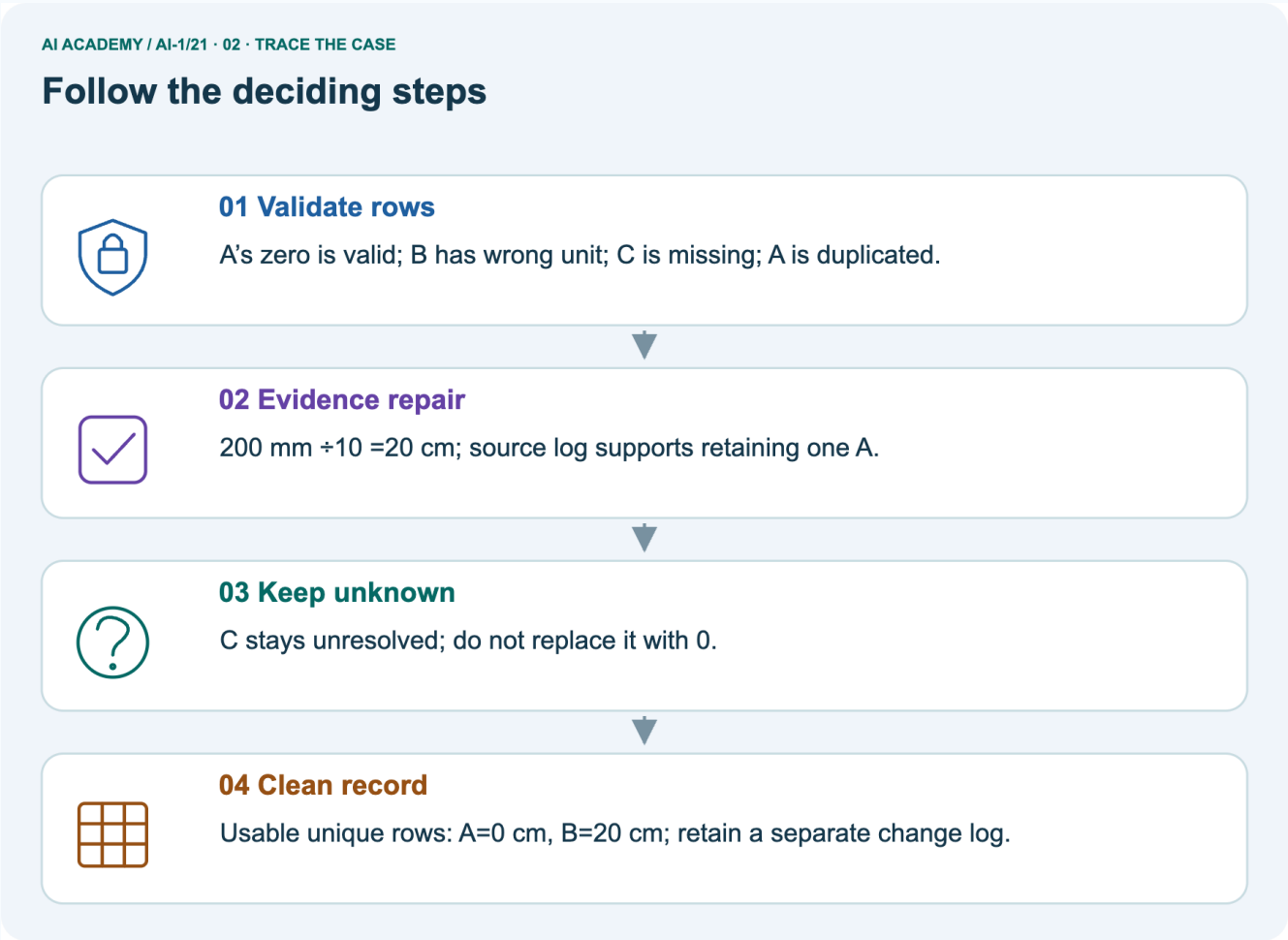
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional toy-car distance table needs cleaning before use. Supplied facts: Contract: unique run ID; whole distance 0–30 cm. Raw rows: A=0 cm, B=200 mm, C=blank, another A=0 cm.

Source log confirms one A run and B's original measurement of 200 mm; C has no measurement.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

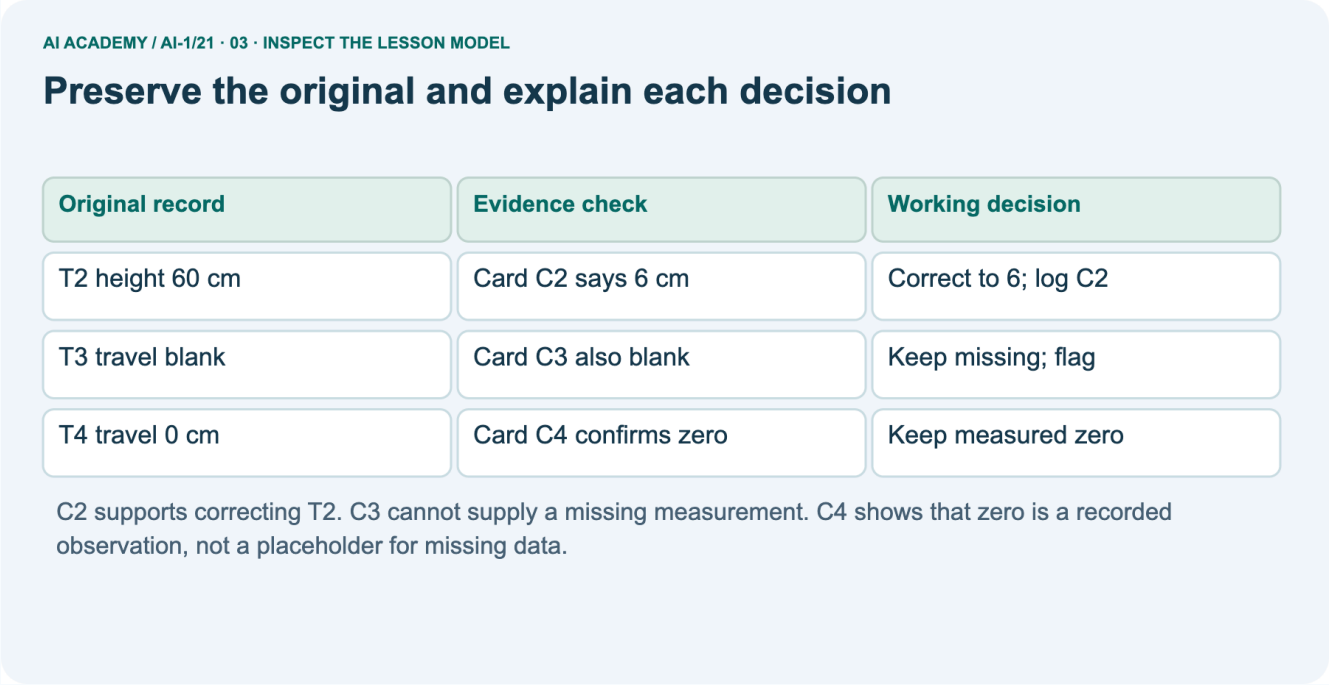


Follow the labelled picture with the matching explanation. Open the vector diagram

Validate rows: A's zero is valid; B has wrong unit; C is missing; A is **duplicate**d. → **Evidence** repair: 200 mm ÷10 =20 cm; source log supports retaining one A. → Keep unknown: C stays unresolved; do not replace it with 0. → Clean record: Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**.

Point to each numbered stage and explain which recorded **input** it uses.

03. Preserve the original and explain each decision



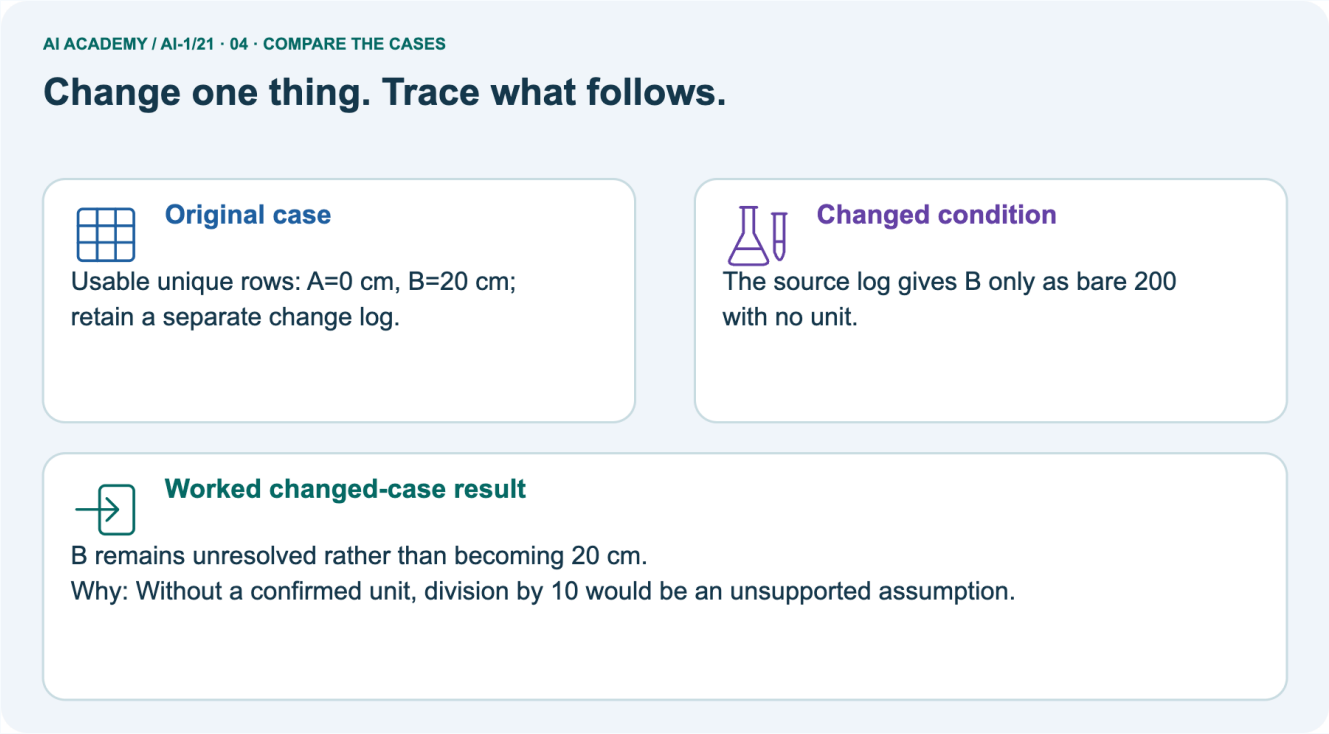
Follow the labelled picture with the matching explanation. Open the vector diagram

Preserve the original and explain each decision. T2 height 60 cm | Card C2 says 6 cm | Correct to 6; log C2. T3 travel blank | Card C3 also blank | Keep missing; flag.

T4 travel 0 cm | Card C4 confirms zero | Keep measured zero

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



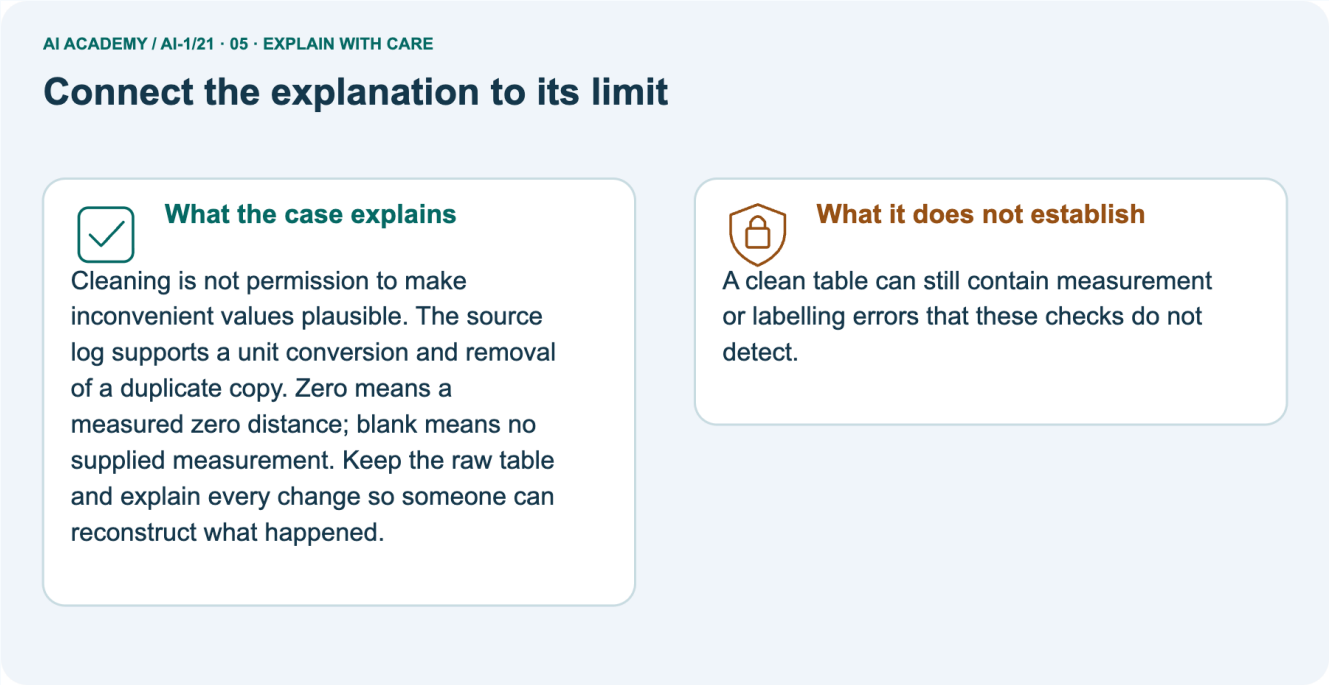
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Usable unique rows: A=0 cm, B=20 cm; retain a separate **change log**. Changed condition: The source log gives B only as bare 200 with no unit. Expected result: B remains unresolved rather than becoming 20 cm.

Reason: Without a confirmed unit, division by 10 would be an unsupported assumption.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



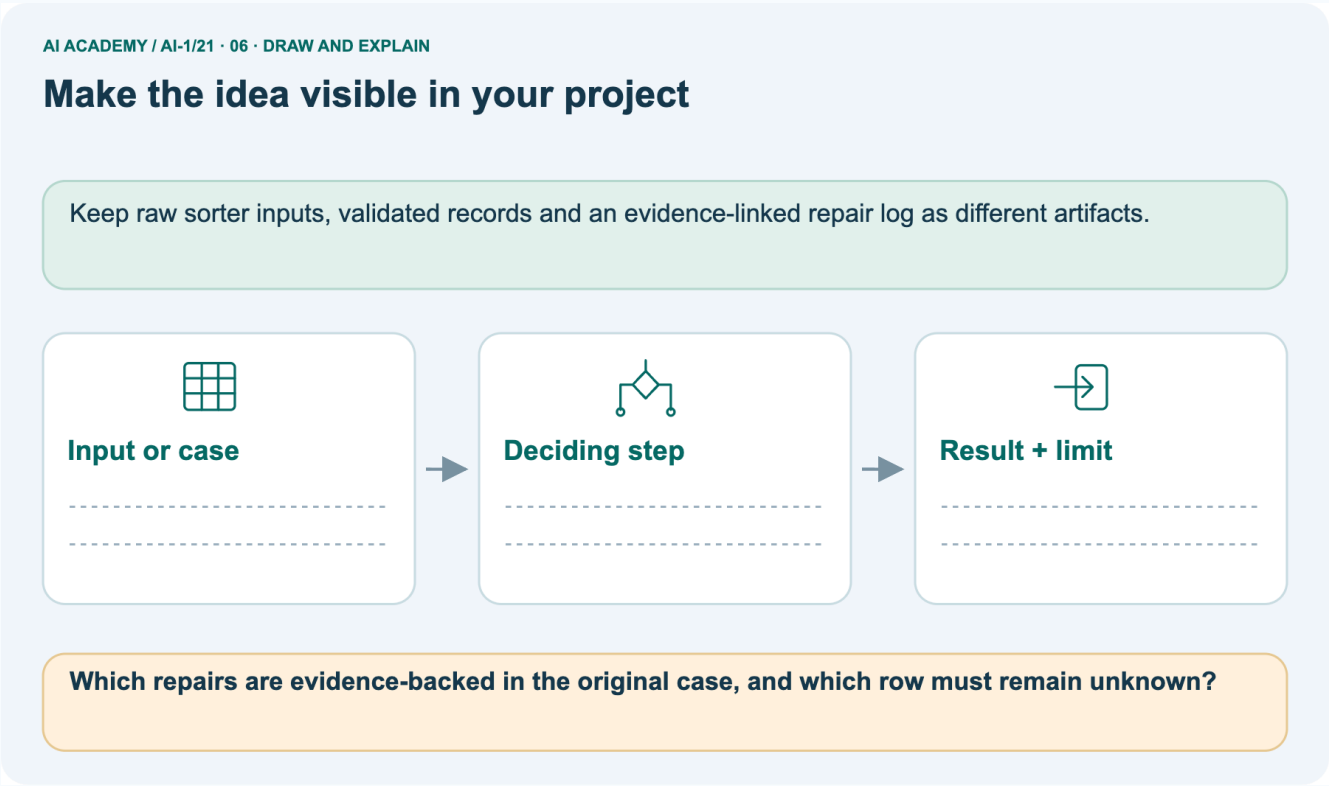
Follow the labelled picture with the matching explanation. Open the vector diagram

Cleaning is not permission to make inconvenient values plausible. The source log supports a unit conversion and removal of a **duplicate** copy. Zero means a measured zero distance; blank means no supplied measurement.

Keep the raw table and explain every change so someone can reconstruct what happened. Limit: A clean table can still contain measurement or labelling errors that these checks do not detect.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep raw sorter **input**s, validated records and an **evidence**-linked repair log as different artifacts. Draw your **input**, deciding step, result and limit.

Supported question: Which repairs are **evidence**-backed in the original case, and which row must remain unknown?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.